

Generating 3D Human Texture from a Single Image with Sampling and Refinement

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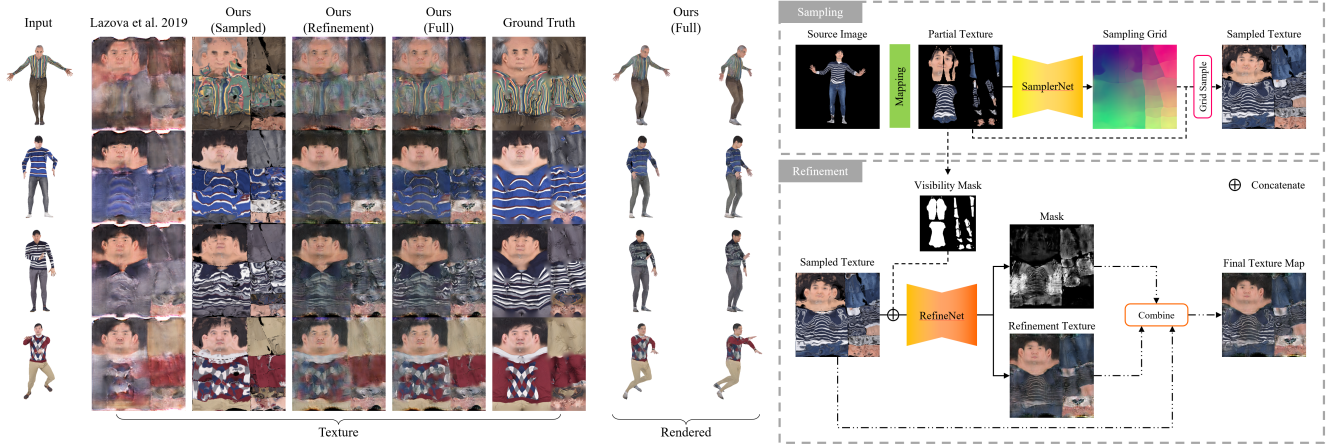


Figure 1: Generated textures and overview of the proposed method.

ABSTRACT

Generating the texture map for a 3D human mesh from a single image is challenging. To generate a plausible texture map, the invisible parts of the texture need to be synthesized with relevance to the visible part and the texture should semantically align to the UV space of the template mesh. To overcome such challenges, we propose a novel method that incorporates SamplerNet and RefineNet. SamplerNet predicts a sampling grid that enables sampling from the given visible texture information, and RefineNet refines the sampled texture to maintain spatial alignment.

CCS CONCEPTS

• **Computing methodologies** → *Computer graphics; Computer vision; Image manipulation.*

KEYWORDS

3D Human Texture Generation, Texture Synthesis

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1 INTRODUCTION

Generating a 3D human avatar from a single image is attracting a lot of attention with the growing popularity of augmented and VR/AR applications. There have been numerous methods proposed to predict a 3D mesh from a single image [Alldieck et al. 2019; Kanazawa et al. 2018]. While the methods can reconstruct the shapes and poses of the 3D mesh, few has studied for estimating a texture map from a single image [Lazova et al. 2019; Wang et al. 2019].

Previous methods for generating texture map from a single image using convolutional neural network (CNN) often produced blurry texture and unable to synthesis the textural pattern of the cloth, where the parts are invisible in the source image. This is a critical because many clothes have repeated patterns.

In this paper, we propose a method to generate a texture map that synthesizes the textural pattern in the invisible parts of the source image. Our method consists of two CNNs: *SamplerNet* and *RefineNet*. *SamplerNet* generates a sampling grid to fill the invisible part of the input partial texture map by referring to the pixels of the visible part. Given the sampled texture map from the sampling grid, *RefineNet* generates a refinement texture map and a mask. The final texture map is produced by alpha blending the sampled and refinement texture map with the predicted mask. We compared our method with a previous method and show that our method outperforms qualitatively and quantitatively.

2 METHOD

Our goal is to produce the full texture map for a 3D human mesh from a single image. We map the human appearance in the source image to the UV space of the SMPL [Loper et al. 2015] using DensePose [Güler et al. 2018] to produce a partial texture map. With the partial texture map as input, our method generates a full texture map through two steps.

Sampler Network. *SamplerNet* predicts a sampling grid based on the image coordinate of the partial texture map to produce a sampled texture map, T_{sample} , where invisible parts are synthesized by referring to visible parts. *SamplerNet* has a U-Net like architecture, in which all layers are residual blocks except for the first layer. *SamplerNet* is trained with the following loss terms:

$$\mathcal{L}_{Sample} = \lambda_{L1}\mathcal{L}_{L1} + \lambda_{LPIPS}\mathcal{L}_{LPIPS}. \quad (1)$$

$\mathcal{L}_{L1}$ and $\mathcal{L}_{LPIPS}$ are L1 loss and perceptual loss [Zhang et al. 2018], respectively. The loss is calculated between T_{sample} and the ground truth texture map. λ_{L1} and λ_{LPIPS} are set to 10 and 1, respectively.

Refine Network. Although DensePose robustly predicts the correspondence between the 2D image and the UV coordinate of the 3D mesh, there remains a slight gap between the predicted coordinates and the actual UV coordinates, producing a misaligned partial texture map. As *SamplerNet* only considers the invisible part of the partial texture map, the visible part remains intact, maintaining misalignment. To alleviate this issue, we use *RefineNet* that produces a refinement texture map T_{refine} and blending mask M . The final output is computed as follows:

$$T_{final} = T_{sample} \odot M + T_{refine} \odot (1 - M), \quad (2)$$

where $\odot$ is hadamard product.

The architecture of *RefineNet* is also based on U-Net, with 9 residual blocks for the bottleneck. *RefineNet* is trained with the following loss terms:

$$\mathcal{L}_{Refine} = \lambda_{L1}\mathcal{L}_{L1} + \lambda_{VGG}\mathcal{L}_{VGG} + \lambda_{FM}\mathcal{L}_{FM} + \lambda_{GAN}\mathcal{L}_{GAN}, \quad (3)$$

where $\mathcal{L}_{VGG}$ and $\mathcal{L}_{FM}$ measure the perceptual difference using the VGG-19 network and a discriminator, respectively. $\mathcal{L}_{GAN}$ is an adversarial loss. λ_{L1} , λ_{VGG} , λ_{FM} , and λ_{GAN} are set to 10, 10, 10, and 1, respectively. All the losses are computed between the ground truth texture map and T_{final} .

3 EXPERIMENTS

Training Details. We used a total of 1,229 texture maps for training and 212 texture maps for testing. We resized the texture maps to 256×256 . We used the Adam optimizer with a learning rate of 0.0002 and beta parameters (0.9, 0.999). The batch size is set to 8, and each model was trained separately for 30,000 iterations on a NVIDIA GTX 1080 Ti GPU.

Evaluation. To evaluate our method, we performed both qualitative and quantitative evaluations. For the quantitative evaluation, we use the Structural Similarity Index Measure (SSIM), Peak Signal-to-Noise Ratio (PSNR), and LPIPS to measure the quality of the generated texture map. We compared the intermediate and final outputs of our method to the final output of [Lazova et al. 2019].

As shown in Table. 1, our method outperforms [Lazova et al. 2019] in all metrics. We can observe that the quality of the final

texture map is improved by adding each network. Additionally, we compared the visual results of the generated textures as shown in Fig. 1. The results from [Lazova et al. 2019] lack detailed patterns and are generally blurry. Our results T_{refine} from *RefineNet* better preserve the details of the input image compared to those from [Lazova et al. 2019]. However, the patterns generated in invisible parts are still blurry. Our final results T_{final} show significant improvement compared to those from [Lazova et al. 2019] and T_{refine} .

Discussion. In this work, we proposed a novel method for generating a 3D human texture map from a single image. Compared to the previous approach, our method generates texture maps with improved quality by synthesizing textural patterns in invisible parts. Our method relies on visible part for the texture synthesis and therefore it is unable to generate non-existing information. An interesting future research direction would incorporate a reference texture to fill out the missing texture regions.

Table 1: Quantitative Comparison.

Method	LPIPS↓	PSNR↑	SSIM↑
[Lazova et al. 2019]	0.2454	17.6744	0.5593
Ours (sampled)	0.2305	17.4234	0.5787
Ours (refinement)	0.2258	17.8700	0.5993
Ours (full)	0.2061	18.1052	0.6032

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